

Federated Single-Index Bandits

Cooperative learning when the reward function itself is unknown

Presented by:

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Course: CS6007: Multi-Agent Machine Learning

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The assumption everybody makes, and why it breaks



Contextual bandits assume we know how context maps to reward:

$$\mathbb{E}[y \mid x] = \underbrace{\langle x, \theta_* \rangle}_{\text{linear bandit}} \quad \text{or} \quad \underbrace{f(\langle x, \theta_* \rangle)}_{\substack{\text{generalized linear} \\ f \text{ known}}}$$

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Dose-response, congestion, click-through and user-satisfaction curves are non-monotone and unknown in advance.

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This is not a mild modelling error

Misspecifying f makes state-of-the-art algorithms incur **linear regret** — they do not degrade gracefully, they fail.¹

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Problem statement: the single-index bandit (SIB)



Model

At round t the learner sees K arms $\mathcal{X}_t = \{x_{t,a}\}_{a \in [K]}$ drawn i.i.d. from a density p on $\mathbb{R}^d$; it pulls x_t and observes

$$y_t = f(\langle x_t, \theta_* \rangle) + \eta_t, \quad \eta_t \text{ } \sigma\text{-sub-Gaussian,}$$

with **both** $\theta_* \in \mathbb{R}^d$ **and** $f : \mathbb{R} \rightarrow \mathbb{R}$ **unknown**. Normalise $\|\theta_*\|_1 = 1$.

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Key tool — Stein's identity, with score $S(x) = -\nabla \log p(x)$

$$\mathbb{E}[y S(x)] = \mathbb{E}[f'(\langle x, \theta_* \rangle)] \theta_* = \mu_* \theta_*$$

The direction is recoverable by a **plain sample average** — with no knowledge of f whatsoever.



Where the literature stands

	unknown link f	multi- agent	safety constr.	optimal rate
Abbasi-Yadkori et al. '11 (OFUL)	✗	✗	✗	—
Chowdhury & Gopalan '17 (IGP-UCB)	✓	✗	✗	—
Dubey & Pentland '20 (Coop-KernelUCB)	✓	✓	✗	—
Amani & Thrampoulidis '21 (Safe-DLUCB)	✗	✓	✓	—
Kang et al. '26 (ESTOR / GSTOR)	✓	✗	✗	✗ $T^{3/4}$
Arya et al. '26 (kernel SIB)	✓	✗	✗	—
Dey, Bhore & Ghosh '26 (ZoomSIB-UCB)	✓	✗	✗	✓ $\tilde{\Theta}(T^{2/3})$
This project (Fed-ZoomSIB)	✓	✓	✓	✓

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Every column has been solved individually.

No work occupies the last row: single-index bandits have never been federated.



Our problem: what does collaboration buy?

Setting

N agents on a connected graph G with communication matrix $\mathbf{P}$. Each round, every agent i sees its own arm set, pulls $x_{i,t}$, observes $y_{i,t} = f(\langle x_{i,t}, \theta_* \rangle) + \eta_{i,t}$. f and θ_* are **shared and unknown**. Agents exchange $\leq B$ values with neighbours every C rounds.

Objective — network regret

$$R_T(N) = \sum_{t=1}^T \sum_{i=1}^N \left[f(\langle x_{i,t}^*, \theta_* \rangle) - f(\langle x_{i,t}, \theta_* \rangle) \right]$$



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How much does cooperation help when the reward function itself must be learned — and at what communication cost?

Why it should work: two phases, two different gains



ZOOMSIB-UCB (Dey, Bhore & Ghosh '26) is two-phase — and the phases federate for *different reasons*.

Phase 1 — estimate θ_*

Truncated Stein estimator over T_0 rounds.

It is a **sample mean** $\Rightarrow$ averaging across agents **equals** the centralized estimator on pooled data.

Exact federation, **one** message of d numbers.

$$T_0 = \tilde{O}\left(\frac{d^2 T^{2/3}}{N}\right)$$

factor N

Phase 2 — UCB over bins

$N_{\text{bins}} = O(T^{1/3})$ bins on the index line; *sleeping* UCB.

Per-bin statistics (n_j, S_j) are two scalars $\Rightarrow$ a standard cooperative MAB over G .

Exploration concentrates only as $\sqrt{\cdot}$:

$$\tilde{O}(\sqrt{N_{\text{bins}} N T})$$

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Prop. 4.7 of Dey et al. is a **modular reduction**: any sleeping-bandit subroutine plugs into Phase 2 — and a cooperative one is exactly that.



Conjectured main theorem

$$\text{Network: } R_T(N) = \tilde{O}(d^2 T^{2/3} + \sqrt{N} T^{2/3})$$

$$\text{Per agent: } R_T(N)/N = \tilde{O}(d^2 T^{2/3}/N + T^{2/3}/\sqrt{N})$$

versus $N \cdot \tilde{O}(d^2 T^{2/3})$ for N independent learners.

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The interesting asymmetry

Estimation cost amortizes **fully** across the network; *exploration* cost only as $\sqrt{N}$.
 $\Rightarrow$ **falsifiable prediction**: collaboration helps *more* in higher dimension —
testable by sweeping d at fixed N .

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Extensions. (a) *Safety*: unknown constraint $h(\langle x, \mu_* \rangle) \leq c$ via a second Stein estimator. (b) *Clustered indices*: M unknown groups $\theta_*^{(m)}$, IFCA-style.



Project plan

Window	Work	Deliverable
29 Sep – 10 Oct	Reproduce single-agent ZOOMSIB-UCB; baselines: Random, LinUCB, ESTOR, GSTOR, IGP-UCB	Regret curve matching Dey et al. Fig. 1
11 – 22 Oct	Fed-ZoomSIB : one-shot Stein averaging + cooperative sleeping-UCB over graph topologies	Speed-up plots; federated concentration lemma
23 – 29 Oct	Communication ablations: interval C , event-triggered, quantization	Regret per transmitted byte
30 Oct – 5 Nov	One extension: safety <i>or</i> clustered indices	Violation-rate / negative-transfer study
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Evaluation: network & per-agent regret; samples to fixed accuracy; messages and bytes; sensitivity to spectral gap $1 - |\lambda_2|$; sweeps over N, d, K , topology, link shape. Synthetic + KDD Cup 99 and Forest Cover Type, as in Dey et al.



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- ④ S. Amani, C. Thrampoulidis. *Decentralized Multi-Agent Linear Bandits with Safety Constraints*. AAAI, 35(8):6627–6635, 2021.
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Thank You!